



**INTELLIGENCE LOAD SHEDDING SYSTEM
MICROCONTROLLERS AND EMBEDDED SYSTEMS
LABORATORY WITH THEORY**

A Report

Submitted by

GOPINATH S	2403412410621024
HARISH KUMAR A	2403412410621026
JAYASURYA R	2403412410621032
UMESH MADHU P	2403412410621074

in partial fulfillment for the award of the

degree of

BACHELOR OF ENGINEERING

in

**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

SRI SAIRAM INSTITUTE OF TECHNOLOGY
(An Autonomous Institution; Affiliated to Anna University, Chennai -600 044)

**ANNA UNIVERSITY: CHENNAI
600044**

APRIL 2026

ABSTRACT

The Intelligence Load Shedding System is designed to prevent the electrical overload by automatically switching the load connections based on priority. In electrical systems, overload conditions can lead to power failure or equipment damage. This project proposes a Microcontroller based solution that continuously monitors the total current consumption and disconnects the non-essential loads based on priority as High, Medium and Low When the demand exceeds the predefined threshold. The system uses a current sensor to measure load consumption and relays to control multiple loads. Each load is assigned a priority level, ensuring that critical loads remain operational while low priority loads are shed during the overload conditions. This approach improves the system reliability, prevents power failure and optimizes the power usage. This project demonstrates the practical applications of embedded systems in power management.

TABLE OF CONTENTS

CHAPTER NO	TITLE	PAGE NO
1	INTRODUCTION	4
2	MOTIVATION FOR THE WORK	4
3	LITERATURE SURVEY	5
4	BLOCK DIAGRAM AND WORKING PRINCIPLE	6-7
5	RESULT AND DISCUSSION	8
6	SCOPE OF THE PROJECT AND CONCLUSION	9
7	REFERENCES	10

CHAPTER 1

Introduction

The rapid increase in electrical power demand has made efficient power management an essential requirement in modern systems. Overloading of electrical systems can lead to equipment damage, power failure, and reduced system reliability. Traditional methods of load management are often manual and inefficient, which may result in improper distribution of available power.

To overcome these issues, an intelligent and automated system is required to monitor and control electrical loads. Microcontroller-based systems provide an effective solution by enabling real-time monitoring and decision-making. This project focuses on designing a smart system that can automatically manage loads based on priority, ensuring efficient utilization of available power and preventing overload conditions.

CHAPTER 2

MOTIVATION FOR THE WORK

The motivation behind this project is to address the problem of electrical overloading and inefficient power distribution. In many systems, there is no proper mechanism to prioritize loads during high demand. This project aims to develop an intelligent system that automatically manages loads based on priority, ensuring efficient and reliable power usage.

CHAPTER 3

LITERATURE SURVEY

The concept of load shedding has been widely studied and implemented in electrical power systems to manage excess demand and maintain system stability. Traditional load shedding methods are generally manual or based on fixed schedules, which often result in inefficient power distribution and unnecessary power interruptions. With advancements in embedded systems, automated load management using microcontrollers has become more feasible and effective. Various studies have focused on using sensors to continuously monitor electrical parameters such as current and voltage, enabling real-time detection of overload conditions.

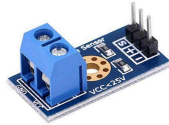
Microcontroller-based systems are preferred due to their simplicity, low cost, and flexibility in operation. Researchers have also introduced priority-based load management techniques, where loads are classified according to their importance, ensuring that critical loads remain active while non-essential loads are disconnected during overload situations. Furthermore, modern smart grid technologies incorporate advanced automated load shedding methods, although they tend to be complex and expensive for small-scale applications. Based on these developments, this project aims to design a simple, cost-effective, and efficient intelligent load shedding system using a microcontroller to monitor load conditions and control power distribution based on priority.

CHAPTER 4

BLOCK DIAGRAM AND WORKING PRINCIPLE



Power supply



Current Sensor(ACS712)



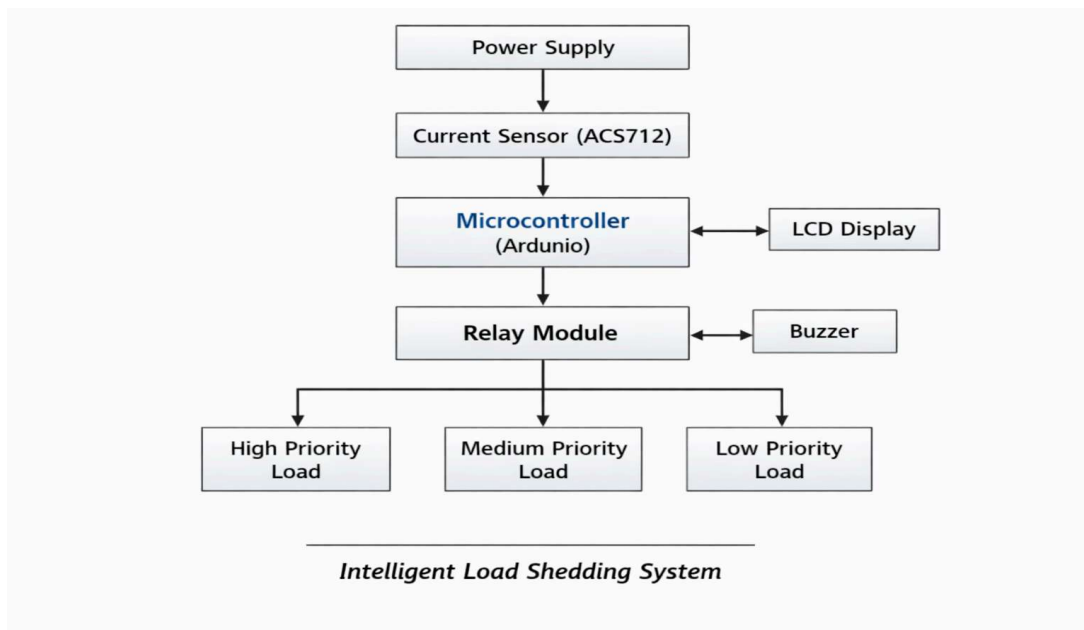
Arduino Uno



Relay Module=(high,medium&low)



Buzzer



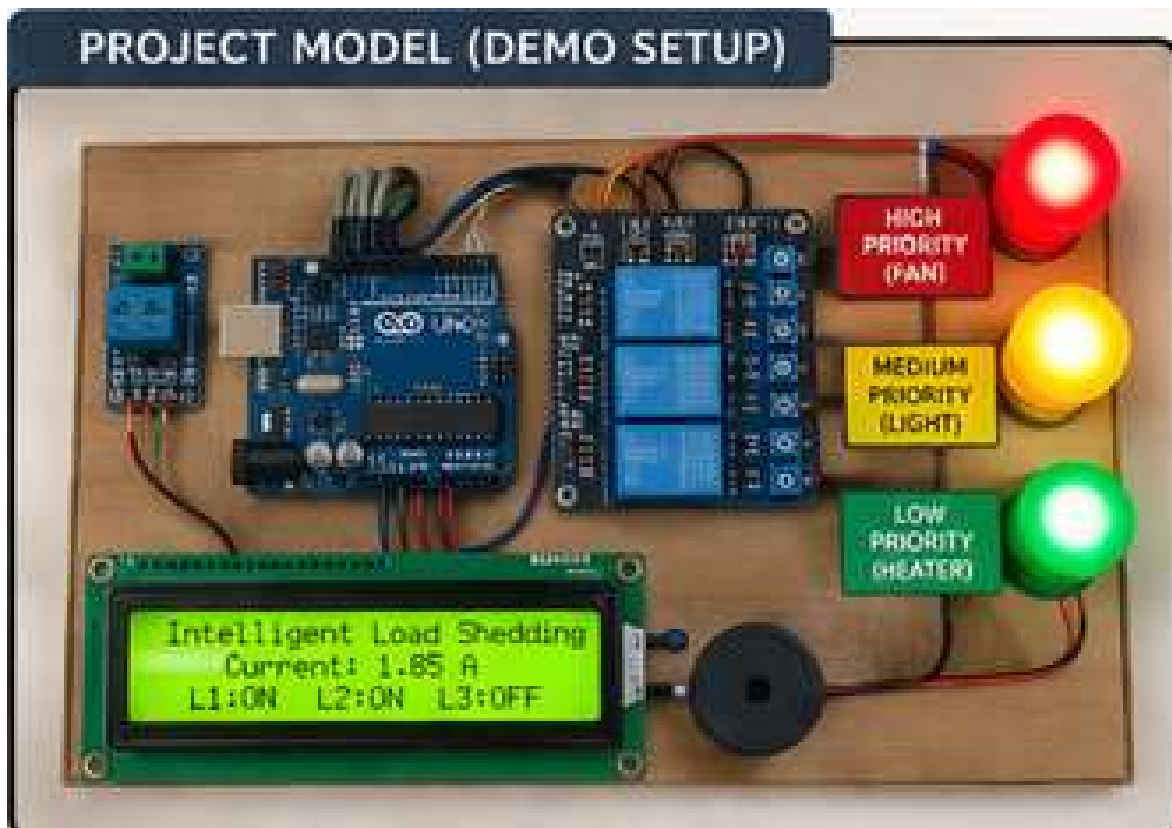
WORKING PRINCIPLE

The Intelligent Load Shedding System operates by continuously monitoring the electrical load using a current sensor. The sensor measures the current flowing through the system and sends the corresponding signal to the microcontroller. The microcontroller processes this data and compares it with predefined threshold values to determine whether the system is operating under normal or overload conditions. Under normal conditions, all connected loads remain active. When the current exceeds the first threshold level, indicating an overload condition, the system automatically disconnects the low-priority load using a relay. If the overload condition persists and the current exceeds a higher threshold, the medium-priority load is also disconnected, while the high-priority load remains active at all times. The relay module acts as a switching device controlled by the microcontroller to turn loads ON or OFF. Additionally, a buzzer provides an alert during overload conditions, and an LCD display shows the current value and load status in real time. This system ensures efficient power management by maintaining supply to critical loads while preventing system failure due to overload.

CHAPTER 5

RESULT AND DISCUSSION

The Intelligent Load Shedding System was successfully implemented and tested. The system monitored the current continuously and detected overload conditions effectively. During normal conditions, all loads remained ON. When the load exceeded the threshold, the low-priority load was disconnected first, followed by the medium-priority load if the overload continued, while the high-priority load remained active. The LCD displayed real-time current and load status, and the buzzer provided an alert during overload. The system performed reliably, demonstrating efficient and automatic load management.



CHAPTER 6

SCOPE OF THE PROJECT

The proposed Intelligent Load Shedding System has wide scope in real-world power management applications. It can be implemented in smart grids to improve efficient power distribution and reduce overload conditions. The system is useful in industries, hospitals, and commercial buildings where uninterrupted power supply to critical loads is essential. It can also be applied in residential systems for better energy management.

The project can be further enhanced by integrating Internet of Things (IoT) technology, enabling remote monitoring and control through mobile or web applications. Advanced features such as automatic data logging, fault detection, and predictive maintenance can also be added. Additionally, the system can be upgraded using more accurate sensors and wireless communication modules for large-scale applications. With further development, it can contribute to energy conservation, improved reliability, and smarter power management systems.

CONCLUSION

The Intelligent Load Shedding System was successfully designed and implemented to manage electrical loads efficiently. The system monitors current in real time and automatically disconnects non-priority loads during overload conditions, ensuring continuous operation of critical loads. It is a simple, cost-effective, and reliable solution for improving power management and system safety.

CHAPTER 10

REFERENCES

1. Muhammad Ali Mazidi, Janice Gillispie Mazidi, Rolin D. McKinlay,
“*The 8051 Microcontroller and Embedded Systems*”, Pearson Education.
2. Simon Monk,
“*Programming Arduino: Getting Started with Sketches*”, McGraw-Hill
Education.
3. Arduino Official Website
<https://www.arduino.cc/>
4. Electronics Hub
<https://www.electronicshub.org/>
5. Circuit Digest
<https://circuitdigest.com/>
6. All About Circuits
<https://www.allaboutcircuits.com/>
7. Texas Instruments
ACS712 Current Sensor Datasheet